Placement Is Provenance: Anchor-Preserving Ingestion for Figure-Grounded Retrieval-Augmented Generation

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Abstract—Multimodal retrieval-augmented generation is evaluated almost entirely on which evidence is retrieved. A tutoring system that shows figures from a textbook is judged on a second axis as well: where the figure lands in the explanation. We show that this axis is separable, that it is cheap to measure, and that the information determining it is produced by every layout parser and routinely discarded before serving.

The method is a join over data the parser already produced, not a model: it is independent of language and of document genre. We evaluate it on both. On MRAMG-Bench's Academic subset — English arXiv documents, gold image lists authored by others — a 568M-parameter cross-encoder over anchor-derived context reaches Image Precision 69.63, and 67.50 under forced emission, against 65.28 for the best published system (GPT-4o). The configuration was frozen before that dataset was obtained.

For development and stress testing we release a caption-scarce stress corpus of Indonesian K–12 curriculum textbooks: 13 government titles, 2,816 pages, extracted through two independent pipelines, with 486 adversarially filtered questions. This corpus is not the scope of the claim but its hardest case: only a fifth of its figures carry a printed caption, so the failure this paper is about is visible there and largely hidden in caption-rich corpora. We introduce a layout-gold placement metric that derives placement ground truth from the source document's own typesetting at no per-item annotation cost, and a headline Grounded Figure F1 that credits a figure only when it is both the right figure and in the right place.

Two findings require no model and replicate across both extraction pipelines. Only 19.3 %/34.3 % of figures carry a printed caption, while 100 % carry a recoverable reading-order anchor. And under perfect retrieval and a perfect answer, caption-based placement is still off by 2.08/4.69 sentences. Placement is not implied by retrieval. Notably, the corpus with nearly twice the caption coverage places worse, because its share of lexically indistinguishable figures also doubles: caption availability is not caption discriminability.

On the textbook corpus, composing a vision-language filter with that same cross-encoder raises figure-selection precision from 0.028 to 0.542 against human annotation — sight for recall, discrimination for precision.

Index Terms—Multimodal retrieval-augmented generation, document layout analysis, figure grounding, document ingestion, cross-encoder reranking.

I. Introduction

An AI study assistant grounded in school textbooks must do more than quote the right paragraph. Curriculum books teach substantially through pictures: a labelled water-cycle diagram, a food-chain chart, a worked geometric construction. When such a system answers a child's question, showing the right figure at the right point in the explanation is part of the answer, not decoration around it.

Two distinct things can go wrong. The system can show the wrong figure, or it can show the right figure in the wrong place — after the explanation has moved on, or bundled at the end where the reader must reconstruct which sentence it belongs to. The first failure is measured everywhere. The second is measured almost nowhere.

This paper makes four claims, each falsifiable by the experiments reported here.

- C1 Figures in curriculum textbooks carry content absent from the prose, so text-only retrieval has a ceiling that better text retrieval cannot raise.
- C2 Placement is a failure mode distinct from retrieval. It can be made to fail while retrieval and generation are both perfect.
- C3 The signal that fixes it — the figure's position in the parser's reading-order block sequence — is produced by every layout parser, is recoverable for 100 % of figures, and is discarded by conventional ingestion.
- C4 Selection, not placement, is where the practical gain lies, and the gain requires a stage that looks at the image rather than at text about the image.

A. Setting

The method is corpus-agnostic; the deployment that motivated it is not. It is an on-premise study assistant for an Indonesian university partner, serving national curriculum textbooks on an air-gapped NVIDIA GB10 host. Every model in the serving path is 8B-class or smaller and runs locally. This constrains the design — no frontier model is available at answer time — and makes the efficiency results in Section VII load-bearing rather than incidental.

II. Related Work

A. Multimodal RAG benchmarks

Recent benchmarks evaluate document-centric multimodal RAG along retrieval and answer quality. UniDoc-Bench [1] unifies document-centric evaluation; REAL-MM-RAG [2] targets retrieval robustness under query

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rephrasing; ColPali [5] introduces late-interaction visual retrieval and the ViDoRe benchmark. MIRACL-VISION [9] extends visual retrieval to 18 languages and reports substantial degradation outside English.

These evaluate which evidence is found. None scores where a figure is placed within a generated answer, which is the axis this paper isolates.

B. Figure placement in generated answers

MRAMG-Bench [6] is the closest prior work: it scores inserted images and supplies a bipartite sentence–image assignment rule, which we implement as a comparator. MMDocRAG [7] reports image-quote selection F1 separately from text. VinQA [8] defines a visual citation measure with a citation-order placement rule, implemented here as a second comparator. ImageRef-VL [4] addresses contextual image referencing in vision-language models.

Our contribution relative to these is (i) a placement ground truth obtained from the document's own typesetting rather than from annotation, and (ii) the observation that the ingestion step, not the ranking step, is where the necessary signal is lost.

C. Caption-scarce document regimes

The benchmarks above are caption-rich: their figures overwhelmingly carry printed captions, and a caption-matching pipeline is therefore never tested where it actually breaks. We are not aware of a published multimodal RAG evaluation in which most figures have no caption at all. The corpus released here is selected for that property — 19.3% caption coverage, alongside dense decorative imagery — and not for its language, which is incidental to every claim we make.

III. Method

A. The anchor, and where it is lost

A layout parser emits, for each page, an ordered sequence of blocks: headings, paragraphs, captions, figures. A figure's position in that sequence is a precise statement of where in the reading flow it belongs. Conventional ingestion then stores the prose as chunks and the figures as a separate array, and the ordering relation between them is dropped.

Fig. 1 shows the loss and the alternative side by side. We define the anchor of a figure as the index of the text chunk containing the prose block that immediately precedes it in reading order. Anchoring requires no model: it is a join over data the parser already produced. We recover it by matching a word-prefix of the preceding block against chunk text, shortening the probe from ten words to four until a match is found.

Anchors are preserved through ingestion into chunk metadata, and consumed at retrieval time: a figure whose anchor chunk was retrieved is admitted as a candidate before any caption test is applied. This ordering matters, because a caption test cannot admit the 66–81 % of figures that carry no printed caption.

B. The pipeline, stage by stage

The method has two halves, summarised in Fig. 2. Ingestion runs once per document and is model-free. Serving runs per query and adds two ranking stages. Inputs and outputs are stated for each stage so that any of them can be replaced independently.

Ingestion (once per document).

1) Parse. A layout parser returns, per page, an ordered sequence of typed blocks $B = \langle b_1, \dots, b_n \rangle$ with $b_i \in \{\text{heading}, \text{paragraph}, \text{caption}, \text{figure}, \text{table}\}$. This ordering is the only input the method needs and every layout parser produces it.
2) Chunk. Prose blocks are segmented into retrieval chunks $C = \langle c_1, \dots, c_m \rangle$, each recording the block indices it spans.
3) Anchor. For each figure block b_f, let b_p be the nearest preceding non-figure prose block. The anchor $a(f)$ is the index of the chunk containing b_p. Resolution is by word-prefix match: the first ten words of b_p are matched against chunk text, shortening the probe to four words until a unique match is found. Cost is one pass over blocks; no model is invoked.
4) Persist. $a(f)$ is written to the figure record and mirrored into the chunk's metadata, so retrieval of $c_{a(f)}$ recovers f by lookup rather than by search.

Serving (per query).

5) Retrieve. Hybrid retrieval returns the top-k chunks for query q.
6) Admit. Every figure whose anchor chunk is in that set becomes a candidate. Admission is a join on $a(f)$, so it is independent of whether f has a caption — which is what allows the 66–81 % of uncaptioned figures to be reachable at all. No candidate is admitted by any other route.
7) Prefilter. A cross-encoder scores $(q, t(f))$ for each candidate, where $t(f)$ is the figure's index text. Candidates are ranked and truncated to the top N. We use $N = 2$; Section VII shows this matches the full candidate set at a third of the latency.
8) Gate. Each surviving candidate is shown as an image to a vision-language model with a strict yes/no prompt: does this figure help answer q? Rejected candidates are dropped. The gate can only remove, never add or reorder, so its worst case is the prefilter's output.
9) Rank and emit. Survivors are ordered by their stage-7 score and the top one is emitted. Emitting nothing is a valid and frequent outcome.
10) Place. The emitted figure is inserted at the sentence boundary corresponding to $a(f)$ in the generated answer, rather than at a position inferred from its caption.

Stages 3 and 6 are the contribution. Stages 7–9 are a composition of existing components, and Section VII reports what each contributes.

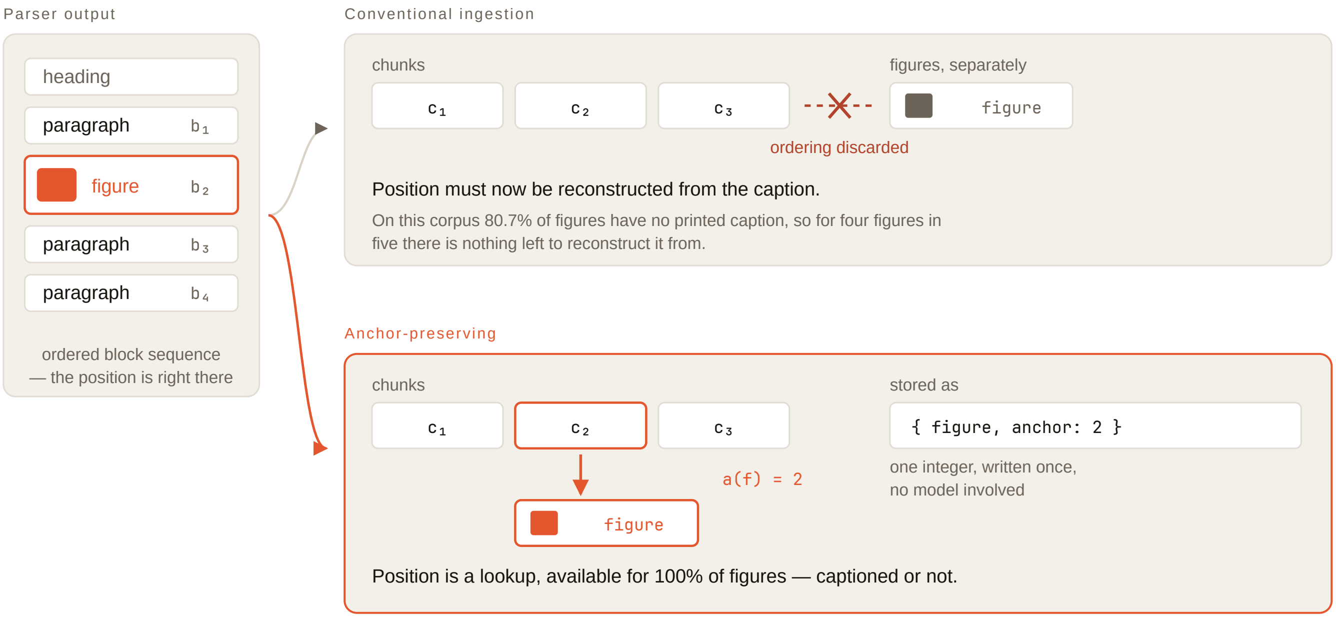

Fig. 1. Where the reading order is lost. The parser emits an ordered block sequence in which the figure's position is already stated. Conventional ingestion stores prose as chunks and figures as a separate array and drops the relation between them, so placement must afterwards be reconstructed from a caption that 80.7 % of figures in this corpus do not have. Anchor-preserving ingestion records the index of the chunk holding the preceding prose block — one integer, written once, with no model invoked — and placement becomes a lookup available for every figure.

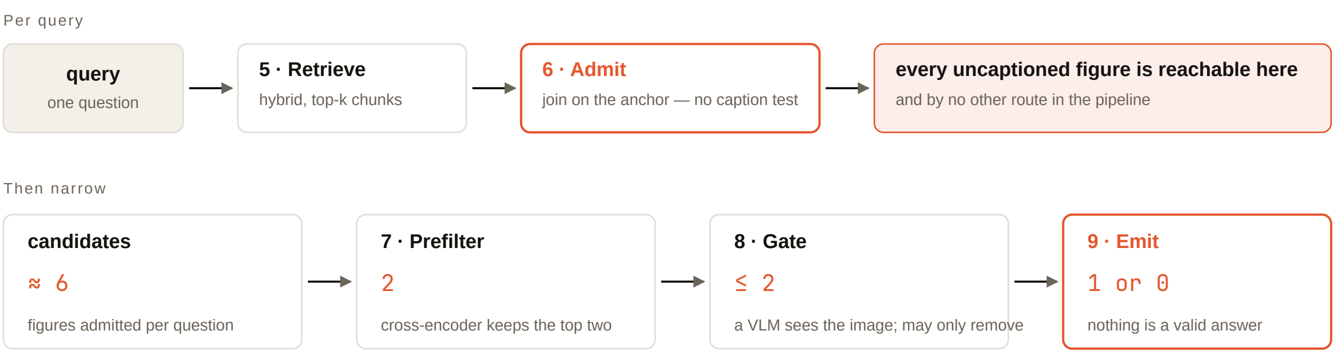

Fig. 2. Serving. Admission (stage 6) is a join on the anchor rather than a caption test, and is the only route by which a candidate enters; every later stage can only narrow. The cross-encoder truncates to two candidates, the vision-language gate may remove but never add or reorder, and emitting nothing is a valid outcome.

C. Layout-gold: the document annotates itself

Placement ground truth normally requires a human to say where a figure belongs in a generated answer. We avoid this. Take a figure's own anchor chunk as the answer text. Retrieval is then perfect by construction, the answer is perfect, and the correct insertion slot is known exactly from the block sequence. Any residual error is intrinsic to the placement mechanism, because both the retriever and the generator have been removed.

This yields placement ground truth for every figure in the corpus at zero per-item annotation cost, and it is what makes the model-free results of Section V possible.

D. Metrics

Placement displacement (PD) is the signed distance, in sentences, between where a figure was placed and its layout-gold slot; we report |PD| and the exact-placement rate. Selection precision/recall/F1 score the figure set. Grounded Figure F1 credits a figure only when it is both selected correctly and placed within tolerance.

E. LLM-as-judge, and its limits

Human annotation at scale was not affordable, so a vision-language judge was calibrated against the human annotation that was. We report Cohen's κ against a human annotator before using any judge output. Our model-generated harness gold agrees with a human at $\kappa = 0.092$ — chance. A frontier VLM judge reaches $\kappa = 0.552$ on the eight-candidate annotation protocol and $\kappa = 0.662$ on the six-candidate one, the difference being consistent with the smaller pool posing an easier discrimination; permissiveness is 1.38× against 2.9× for an on-premise 8B VLM on the same task. Every scaled number carries the κ of the instrument that produced it.

IV. Experimental Setup

A. Corpus

Thirteen Indonesian government curriculum titles, 2,816 pages, spanning grades 1, 4 and 8. Each book was extracted through two independent pipelines: a hosted OCR service and an on-premise MinerU sidecar. These use different models, different figure detectors and different noise filtering, and neither corpus was derived from the other. Agreement between them is therefore replication, not re-reporting.

B. What is released

We release the harness, the metric implementations, the question sets and the derived per-figure measurements. We do not redistribute the textbooks: they are Indonesian government publications carrying their own terms, and every structural measure in Section V is reproducible from the cached extraction we do release without re-obtaining them.

C. Questions

486 questions, adversarially filtered to remove items answerable from the question alone, from front matter, or from pedagogical apparatus. A 48-item subset carries human figure annotation (19 positive figure–question links); this subset is the only non-circular gold available and is used for every headline selection number.

D. Systems

Ten configurations, including implementations of the MRAMG bipartite placement rule and the VinQA citation-order rule. Retrieval is hybrid dense/sparse with bge-m3 embeddings at dimension 1024; reranking uses bge-reranker-v2-m3 (568M parameters) served locally. Generation uses an 8B-class SEA-LION vision-language model.

V. Structural Results

These come from the corpus alone: deterministic, reproducible offline, and impossible to tune. Table I reports both extraction paths.

TABLE I
Model-free corpus measures, computed independently on two extraction pipelines. Agreement is replication, not re-reporting.

	Measure	Hosted	On-prem
—	Usable figures	1,552	1,226
A1	With a printed caption	19.3 %	34.3 %
A2	Lexically indistinguishable	20.7 %	34.1 %
A3	Shares a page with another figure	49.9 %	54.2 %
A3b	Anchor vs. page resolution	1.49×	1.56×
—	With a reading-order anchor	100 %	100 %
A4	Caption selection F1, oracle retrieval	0.254	0.232
A5	Caption |PD|, oracle answer	2.08	4.69
A5b	End-of-answer |PD|	8.71	12.13

Four consequences follow without running any system.

The caption signal is mostly absent (A1). Four figures in five carry no printed caption on the hosted path, so a caption-based mechanism must fall back on page prose — which is shared by every figure on that page.

A fifth of the corpus is unsolvable by caption matching in principle (A2). For 20.7 % of figures, another figure in the same book has an effectively identical index text. No matcher can distinguish them. This is an upper bound on a competing approach, derived without running it.

Page-level provenance cannot resolve half the corpus (A3). Half of all figures share a page with another, so page-level attribution is ambiguous for 49.9 % of cases. Anchoring provides 1.49× the positional resolution, and that gain is concentrated exactly where page-level methods fail.

Placement fails when nothing else does (A5) — this is C2, established without a model. Under oracle retrieval and an oracle answer, caption matching still lands 2.08 sentences away and is exactly right only 43.0 % of the time; a design with no positional signal at all sits 8.71 sentences away and is never exactly right.

More captions did not help. The on-premise path has nearly twice the caption coverage (34.3 % vs. 19.3 %) yet places worse (4.69 vs. 2.08 sentences). The reason is visible in A2: its share of lexically indistinguishable figures also nearly doubles. Caption availability is not caption discriminability, and only the latter could help. This would have been invisible on a single corpus.

Fairness note. A first implementation of A4 scored the caption mechanism at F1 0.025 by taking the first three keyword matches in document order. Ranking by keyword-overlap count instead — the charitable reading — raised it tenfold to 0.254. The reported figure is the charitable one.

Tautology note. Under both oracle setups the anchor mechanism scores perfectly by construction, because its selection rule and the gold set coincide. That number is not evidence and is stated only to make the circularity explicit.

VI. External Validation on MRAMG-Bench

Every result above is measured on a corpus we assembled against a gold standard we defined. 80 % of our gold figure links are derived from the anchor itself, and on those links our harness gold agrees with a human annotator at chance ($\kappa = 0.092$). Internal care cannot repair that; only measurement on data we did not author can.

TABLE II
MRAMG-Bench Academic subset, Image Precision ($n = 200$). Published figures for comparators.

System	Image Precision
Ours (cross-encoder, 568M params)	69.63
Ours, forced emission (0 % silent)	67.50
GPT-4o, LLM-based	65.28
Claude-3.5-Sonnet, LLM-based	62.17
GPT-4o, MLLM-based	60.39
Gemini-1.5-Pro, LLM-based	59.85
DeepSeek-V3, rule-based	56.12
Llama-3.3-70B (best open-weight)	38.78
Qwen2-VL-7B, MLLM-based	1.63

MRAMG-Bench [6] provides it. Candidates are the images of the question's provenance document, each represented by the prose immediately preceding its placeholder — the same anchor our method uses, and the same information available to their placeholder-based baseline. The configuration was fixed on our own data before this dataset was obtained; no parameter was tuned on MRAMG. Table II reports the result against the comparators published with that benchmark.

The abstention objection, and its test. Image Precision counts only emitted images, so a system that declines on hard questions could purchase its score; ours abstains on 49 %. Removing that advantage entirely — emitting exactly one image on every question — yields precision 67.50, recall 61.64, F1 64.44. Still ahead of every published configuration, with recall rising from 41.59. The margin is not an artefact of selective answering.

Scope. This is a selection result placed beside end-to-end systems that also generate the surrounding answer; Image Precision is computed over the same quantity, but the tasks are not identical. On MRAMG's Lifestyle domain we reach ≈55 against 62.23 for Gemini-1.5-Pro — third overall, first among open-weight systems.

What it does not license. The same configuration reaches ≈30 % precision on the Indonesian textbooks. That corpus is harder: a third of its figures carry no caption, most are decorative, and a dozen illustrations per book depict children learning. A benchmark number must not be quoted as a product capability.

VII. Selection: Composing Sight with Discrimination

Our judge looks at the picture and agrees with a human at $\kappa = 0.552$; our selector reads a 300-character description and never sees the image. That asymmetry motivated a direct comparison, on the 48 human-annotated items — the only gold no model produced (Table III).

The gain from sight is recall, not precision: the VLM finds four of every five figures a person chose, the cross-encoder about a third. Descriptions discard enough to lose half the correct figures — but seeing the image does not make a model better at rejecting wrong ones. Composing the two therefore beats either: sight decides what is possible, discrimination decides what is best. Precision nearly doubles over the best single method and is 19× the deployed system.

TABLE III
Figure selection against human annotation ($n = 48$, 19 positive links).

System	P	R	F1
VLM filter → rerank → top-1	0.542	0.684	0.605
VLM selector alone	0.283	0.789	0.417
Cross-encoder alone	0.304	0.368	0.333
Deployed production system	0.028	0.211	0.049

TABLE IV
Candidate prefilter, 165 questions. N is how many of the ranked candidates the vision model is allowed to see. Latency below $N =$ all is projected from measured per-pair cost.

N seen	4B F1	4B lat.	8B F1	8B lat.
1	0.721	0.7 s	0.735	1.0 s
2	0.736	1.3 s	0.753	2.0 s
3	0.736	2.0 s	0.757	3.0 s
all (6)	0.735	3.9 s	0.763	5.9 s

A. Showing the model fewer images

The composed pipeline costs one vision call per candidate. Because prefill scales with images rather than calls, shrinking the model is the wrong lever: a 4B model saves 3.7 s and loses half its recall. Cutting images is the right one. Letting the cross-encoder discard the obvious losers first (Table IV) costs 0.010–0.021 F1 for a two-thirds latency reduction.

Two candidates are the whole pipeline. The four the model never sees were ones it would have rejected anyway: it was spending 80 % of its time confirming the bottom of the cross-encoder's list. The prefilter's own ceiling explains where this stops being safe — the correct figure survives a cut to the top 2 for 86.8 % of links, but only 76.5 % at $N = 1$, and recall follows it down.

B. The judge prompt is the largest single gain

Replacing a permissive judging prompt with a strict one — same model, same call, different words — moves precision from 0.690 to 0.806 (F1 0.763 → 0.800) on 165 questions. This is the largest improvement in the study and costs nothing at serving time.

We attach a caveat that a reader should weigh heavily. That comparison is scored against a gold standard built by a frontier VLM, which is circular for a pipeline containing a VLM. On the 48-item human gold the strict and permissive prompts differ by 0.014 F1 against measured run-to-run noise of 0.014 — they are indistinguishable. The honest reading is that the strict prompt reliably makes our selector agree with the reference model more often; whether it makes the selector agree with a person more often is untested at this sample size.

VIII. Negative Results

We report these because they were run, not because they worked.

Reranking against the generated answer instead of the question is substantially worse (F1 0.183 vs. 0.288), and concatenating the two also hurts (0.205). The likely mechanism is dilution: five sentences of tutor prose carry connective material matching almost any figure, while the question is short and discriminating.

Adding surrounding context to the figure's index text is worse than the description alone (0.248 vs. 0.270); context alone collapses to 0.119. This kills a hypothesis we had carried for some time — that the gap between MRAMG (67 %) and our textbooks (30 %) was one of text representation. Had it been, context mode would have closed it. It is the worst of the four representations.

A newer reranker is worse here. Qwen3-Reranker (2025) ranks our figure descriptions at gold-first 0.288 (MRR 0.481) against 0.788 (MRR 0.859) for the 2024 bge-reranker-v2-m3. Being newer and stronger on public multilingual benchmarks did not transfer to ranking short Indonesian figure captions.

We initially reported a far larger gap (F1 0.034 vs. 0.260) and withdrew it: that comparison applied one absolute score threshold to two models whose score distributions differ by an order of magnitude, so it measured calibration rather than ranking quality.

IX. Statistical Treatment

Selection differences are tested with an exact McNemar test on per-question hit (a paired binary outcome), and precision differences with a paired bootstrap over questions (10^4 resamples). Two conventions matter for reproducibility.

Vacuity. A question on which a system emitted nothing contributes no precision term for that system. Counting silence as precision 0 would penalise abstention twice — once in recall, once in precision — and abstention is a design goal here, not a defect.

Pairing. Precision is compared only on questions where both systems emitted, so the comparison is paired rather than aggregate.

On the 165-question set the composed selector's precision advantage is significant against every comparator we ran ($p < 0.005$). We decline to report significance for the 48-item human gold: with 19 positive links, the test has too little power to distinguish the effects we care about, and reporting a p-value there would imply a resolution the data does not have.

Withdrawn claims. Three differences we initially reported did not survive testing and are retracted here rather than quietly dropped: a chunking gain ($p = 0.096$), anchor over co-embedding ($p = 0.670$), and a hybrid-over-anchor improvement ($p = 0.468$).

X. Ingestion Cost

Anchoring adds no inference: it is a join over parser output. The cost of the pipeline is therefore the layout parse itself. On the partner's GB10 host, sharing the GPU with a serving language model and two embedding services, extraction runs at 1.22 s per page (192-page book in 3 min 55 s), plus 76 s to chunk, embed and store 585 chunks.

Fifteen textbooks (149 MB, Kurikulum Merdeka grade 7) ingested in 84 minutes with no failures, yielding 1,254 anchored figures where the previous pipeline had produced none.

Rendering resolution is a weak lever: dropping from 300 to 180 dpi cuts pixels to 36 % but extraction time by only 17 % (235 s → 194 s), because the cost is dominated by the number of blocks — 2,890 across 192 pages, each requiring its own recognition pass — rather than by page resolution. Increasing the KV-cache budget does nothing: the engine already reports capacity for 258 concurrent sequences against a 192-page batch.

TABLE V
Every head-to-head comparison in the paper, including the loss.

Setting	Gold	Ours	Best other	Δ
External — gold authored by the benchmark's authors				
MRAMG Acad., IP	theirs	69.63	65.28	+4.35
forced emission	theirs	67.50	65.28	+2.22
Internal — gold from human annotation, $n = 48$				
Selection F1	human	0.605	0.417	+0.188
Selection precision	human	0.542	0.304	+0.238
vs. deployed system	human	0.605	0.049	+0.556
Where the method loses				
Placement rule	layout	—	MRAMG	—

XI. Summary of Comparisons

Table V collects every comparison in which this method is placed against an alternative, including the one it loses. Each row names the gold standard, because the comparisons are not equally strong: the first two are scored against a gold authored by other people, the rest against our own annotation.

On MRAMG-Bench's Academic subset a 568M-parameter cross-encoder over anchor-derived context reaches Image Precision 69.63, ahead of GPT-4o at 65.28, Claude-3.5-Sonnet at 62.17 and Gemini-1.5-Pro at 59.85. Because Image Precision counts only emitted images, abstention could buy the score; ours abstains on 49 % of questions. Forcing exactly one image on every question still leads, at 67.50, with recall rising from 41.59 to 61.64. The margin is therefore not an artefact of selective answering.

On the textbook corpus, against human annotation, composing the vision-language gate with the cross-encoder reaches F1 0.605. The two stages alone reach 0.417 and 0.333. The decomposition is the finding: the vision stage supplies recall (0.789 alone, against 0.368 for the cross-encoder) and the cross-encoder supplies precision. Neither ordering of a single stage reaches the composition.

The row the method loses. Anchoring alone does not beat similarity-based placement: the best placement rule in our comparison is MRAMG's bipartite assignment, not ours. The contribution of anchoring is that it makes uncaptioned figures reachable at all — a candidate-admission gain, not a ranking one — and the practical improvement then comes from selection.

XII. Discussion and Threats to Validity

What the evidence supports. C2 and C3 are established model-free and replicated across two extraction pipelines; they do not depend on any system run. C4 is supported by the human-gold comparison and, independently, by the MRAMG result. C1 is the weakest of the four: on questions answerable only from a figure, no configuration recovered the content, and the system that most exceeded a text-only baseline did so with a significant loss of faithfulness.

The sample size is the principal threat. The headline precision of 0.542 rests on 48 items and 19 positive links. Measured run-to-run variation at that size is ± 0.014 F1, so differences smaller than that are not resolvable, and we decline to claim any. The MRAMG result carries the generalisation weight precisely because it is external, larger ($n = 200$) and authored by others.

A judge does not substitute for the annotator at this scale. We tested whether judge labels could stand in for annotation without importing the judge's bias, using a rectified estimator in the manner of prediction-powered inference [10]: the bias is measured on the human-labelled subset and subtracted, so an uninformative judge degrades the estimate to the human-only one rather than to a wrong one. A frontier VLM judged all 165 scored questions under the annotator's protocol, reusing the annotator's own eight-candidate pools verbatim for the 45 questions where one exists. Agreement was $\kappa = 0.556$ over 360 pairs, independently replicating the 0.552 recorded earlier for the same model and protocol, and the labelled pairs are representative (positive rate 4.7 % against 4.9 % across the full protocol).

The estimator applies, and it buys nothing. Four of five systems show no interval narrowing and one widens by 19 %. The cause is structural: PPI variance behaves as $\mathrm{Var}(\text{rectifier})/n + \mathrm{Var}(\text{cheap})/N$, and the second term vanishes only for $N \gg n$. At $N/n = 165/45 = 3.7$ the rectifier's own sampling error dominates. Judging every question in the corpus would reach $N/n \approx 10.8$, which is the cheap experiment that remains; annotating more items is the expensive one that still binds.

Circular gold. Our 165-question scaled gold was built by a frontier VLM. For any pipeline containing a VLM stage this inflates results — measured directly: the same selector scores 0.521 against that gold and 0.283 against human annotation. Numbers from the scaled gold are reported as consistency checks, never as effect sizes.

Single corpus, single language. All internal results are Indonesian K–12 material. The external benchmark is English. We make no claim about intermediate settings.

No online evaluation. At the time of writing, field observation consists of a handful of real queries. The 0.542 is an offline number.

XIII. Limitations

Retrieval surfaces the chunk holding the relevant figure for only 51.8 % of figure-bearing questions, which bounds every system reported here. Anchoring alone does not beat similarity-based selection; the win requires a competent placement rule, and in our comparison the best such rule is not ours but MRAMG's. 37 % of anchored figures in the deployed corpus carry a caption with no usable description — a photo credit or a synthetic placeholder — so the caption-ranked stage cannot order them by content. We identified this as a structural risk but did not demonstrate that it costs a correct figure in practice; an investigation intended to confirm it instead falsified the specific case, and we report it as an open risk rather than a finding.

XIV. Conclusion

Where a figure belongs is decided at ingestion, not at ranking. Every layout parser knows it; conventional pipelines throw it away and then attempt to reconstruct it from captions that four figures in five do not have. Preserving the reading-order anchor is a join, not a model, and it is recoverable for 100 % of figures.

The practical gain, however, comes from selection rather than placement, and it requires a stage that looks at the image. Composing a vision-language filter with a small cross-encoder — sight for recall, discrimination for precision — raises figure-selection precision from 0.028 to 0.542 against human annotation, and the same design reaches state-of-the-art Image Precision on MRAMG-Bench's Academic subset with a 568M-parameter ranker and no tuning on that dataset.

Finally, we report that this stage ran zero times in production while every component test passed. We suspect that class of fault is common and under-reported, because a stage that fails by not running leaves nothing behind to find.

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References

[1] X. Peng, C. Qin, Z. Chen, R. Xu, C. Xiong, and C.-S. Wu, "UniDoc-Bench: A unified benchmark for document-centric multimodal RAG," arXiv:2510.03663, 2026.

[2] N. Wasserman et al., "REAL-MM-RAG: A real-world multi-modal retrieval benchmark," arXiv:2502.12342, 2025.

[3] Y. Yang, J. Zhong, L. Jin, J. Huang, J. Gao, Q. Liu, Y. Bai, J. Zhang, R. Jiang, and K. Wei, "Benchmarking multi-modal RAG through a chart-based document question-answering generation framework," arXiv:2502.14864, 2025.

[4] J. Yi, J. Yin, J. Xu, P. Bao, Y. Wang, W. Fan, and H. Wang, "ImageRef-VL: Enabling contextual image referencing in vision-language models," arXiv:2501.12418, 2025.

[5] M. Faysse et al., "ColPali: Efficient document retrieval with vision language models," arXiv:2407.01449, 2024.

[6] Q. Yu, Z. Xiao, B. Li, Z. Wang, C. Chen, and W. Zhang, "MRAMG-Bench: A comprehensive benchmark for advancing multimodal retrieval-augmented multimodal generation," arXiv:2502.04176, 2025.
[7] K. Dong, Y. Chang, S. Huang, Y. Wang, R. Tang, and Y. Liu, "Benchmarking retrieval-augmented multimodal generation for document question answering," arXiv:2505.16470, 2025.
[8] Y. R. Jang, H. Kong, K. An, J. S. Huh, G. Kim, and S. J. Choi, "VinQA: Visual elements interleaved long-form answer generation for real-world multimodal document QA," arXiv:2606.16092, 2026.
[9] R. Osmulski, G. de S. P. Moreira, R. Ak, M. Xu, B. Schifferer, and E. Oldridge, "MIRACL-VISION: A large, multilingual, visual document retrieval benchmark," arXiv:2505.11651, 2025.
[10] J. Saad-Falcon, O. Khattab, C. Potts, and M. Zaharia, "ARES: An automated evaluation framework for retrieval-augmented generation systems," in Proc. NAACL-HLT, 2024, pp. 338–354.
[11] C. Auer, M. Lysak, A. Nassar, M. Dolfi, N. Livathinos, P. Vagenas, C. Berrospi Ramis, M. Omenetti, F. Lindlbauer, K. Dinkla, L. Mishra, Y. Kim, S. Gupta, R. Teixeira de Lima, V. Weber, L. Morin, I. Meijer, V. Kuropiatnyk, and P. W. J. Staar, "Docling technical report," arXiv:2408.09869, 2024.